

ME315A_FarahLab07

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Lab 07

1. Baixe o arquivo uwmadison.sqlite3. Conecte-se a ele usando o pacote RSQLite, armazenando a conexão em uma variável conn.

```
library(DBI)
library(RSQLite)

conn <- dbConnect(SQLite(), "database.sqlite3")
```

```
dbListTables(conn)
```

```
[1] "course_offerings"  "courses"           "grade_distributions"
[4] "instructors"       "rooms"              "schedules"
[7] "sections"          "subject_memberships" "subjects"
[10] "teachings"
```

```
dbListFields(conn, "subjects")
```

```
[1] "code"      "name"      "abbreviation"
```

```
dbListFields(conn, "subject_memberships")
```

```
[1] "subject_code"      "course_offering_uuid"
```

```
dbListFields(conn, "course_offerings")
```

```
[1] "uuid"      "course_uuid" "term_code"  "name"
```

```
dbListFields(conn, "sections")
```

```
[1] "uuid"          "course_offering_uuid" "section_type"  
[4] "number"        "room_uuid"          "schedule_uuid"
```

```
dbListFields(conn, "teachings")
```

```
[1] "instructor_id" "section_uuid"
```

```
dbListFields(conn, "instructors")
```

```
[1] "id"    "name"
```

```
dbListFields(conn, "grade_distributions")
```

```
[1] "course_offering_uuid" "section_number"    "n_count"  
[4] "c_count"             "f_count"           "cr_count"  
[7] "p_count"             "nw_count"          "other_count"  
[10] "bc_count"            "u_count"           "d_count"  
[13] "ab_count"            "a_count"           "s_count"  
[16] "i_count"             "b_count"           "nr_count"
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (subjects.abbreviation='STAT');

```
SELECT DISTINCT instructors.id, instructors.name  
FROM instructors  
JOIN teachings ON teachings.instructor_id = instructors.id  
JOIN sections ON sections.uuid = teachings.section_uuid  
JOIN course_offerings ON course_offerings.uuid = sections.course_offering_uuid  
JOIN subject_memberships ON subject_memberships.course_offering_uuid = course_offerings.uuid  
JOIN subjects ON subjects.code = subject_memberships.subject_code  
WHERE subjects.abbreviation = 'STAT'  
ORDER BY instructors.name
```

Table 1: Displaying records 1 - 10

id	name
4787911	ABIGAIL BENZINE
3111842	ADIN-CRISTIAN ANDREI
3775809	AKICHIKA OZEKI
4958796	ALAN HUANG
5451706	ALBERTO DEL PIA
3661782	ALBRECHT KLEMM
5656034	ALEXANDER COVINGTON
4680082	ALEXANDER FISH
3573775	ALEXANDER KISELEV
5162339	ALYSSA DIGILIO

3. O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

A) Quem é o professor mais difícil?

```
SELECT instructors.name,
       SUM(grade_distributions.a_count*4 + grade_distributions.ab_count*3.5 +
           grade_distributions.b_count*3 + grade_distributions.bc_count*2.5 +
           grade_distributions.c_count*2 + grade_distributions.d_count*1) * 1.0 /
       SUM(grade_distributions.a_count + grade_distributions.ab_count +
           grade_distributions.b_count + grade_distributions.bc_count +
           grade_distributions.c_count + grade_distributions.d_count +
           grade_distributions.f_count) AS gpa_medio,
       SUM(grade_distributions.a_count + grade_distributions.ab_count +
           grade_distributions.b_count + grade_distributions.bc_count +
           grade_distributions.c_count + grade_distributions.d_count +
           grade_distributions.f_count) AS total_alunos
FROM instructors
JOIN teachings ON teachings.instructor_id = instructors.id
JOIN sections ON sections.uuid = teachings.section_uuid
JOIN course_offerings ON course_offerings.uuid = sections.course_offering_uuid
JOIN subject_memberships ON subject_memberships.course_offering_uuid = course_offerings.uuid
JOIN subjects ON subjects.code = subject_memberships.subject_code
JOIN grade_distributions ON grade_distributions.course_offering_uuid = course_offerings.uuid
                        AND grade_distributions.section_number = sections.number
WHERE subjects.abbreviation = 'STAT'
```

```

GROUP BY instructors.id, instructors.name
HAVING total_alunos > 0
ORDER BY gpa_medio ASC
LIMIT 1

```

Table 2: 1 records

name	gpa_medio	total_alunos
JAMES D KUELBS	2.6375	40

B) Quem é o professor mais fácil?

```

SELECT instructors.name,
    SUM(grade_distributions.a_count*4 + grade_distributions.ab_count*3.5 +
        grade_distributions.b_count*3 + grade_distributions.bc_count*2.5 +
        grade_distributions.c_count*2 + grade_distributions.d_count*1) * 1.0 /
    SUM(grade_distributions.a_count + grade_distributions.ab_count +
        grade_distributions.b_count + grade_distributions.bc_count +
        grade_distributions.c_count + grade_distributions.d_count +
        grade_distributions.f_count) AS gpa_medio,
    SUM(grade_distributions.a_count + grade_distributions.ab_count +
        grade_distributions.b_count + grade_distributions.bc_count +
        grade_distributions.c_count + grade_distributions.d_count +
        grade_distributions.f_count) AS total_alunos
FROM instructors
JOIN teachings ON teachings.instructor_id = instructors.id
JOIN sections ON sections.uuid = teachings.section_uuid
JOIN course_offerings ON course_offerings.uuid = sections.course_offering_uuid
JOIN subject_memberships ON subject_memberships.course_offering_uuid = course_offerings.uuid
JOIN subjects ON subjects.code = subject_memberships.subject_code
JOIN grade_distributions ON grade_distributions.course_offering_uuid = course_offerings.uuid
                        AND grade_distributions.section_number = sections.number
WHERE subjects.abbreviation = 'STAT'
GROUP BY instructors.id, instructors.name
HAVING total_alunos > 0
ORDER BY gpa_medio DESC
LIMIT 1

```

Table 3: 1 records

name	gpa_medio	total_alunos
SRIKANTHMADHAVAN ARAVAMUTHAN	4	16

C) Qual é a disciplina mais difícil?

```

SELECT courses.name,
    SUM(grade_distributions.a_count*4 + grade_distributions.ab_count*3.5 +
        grade_distributions.b_count*3 + grade_distributions.bc_count*2.5 +
        grade_distributions.c_count*2 + grade_distributions.d_count*1) * 1.0 /
    SUM(grade_distributions.a_count + grade_distributions.ab_count +
        grade_distributions.b_count + grade_distributions.bc_count +
        grade_distributions.c_count + grade_distributions.d_count +
        grade_distributions.f_count) AS gpa_medio,
    SUM(grade_distributions.a_count + grade_distributions.ab_count +
        grade_distributions.b_count + grade_distributions.bc_count +
        grade_distributions.c_count + grade_distributions.d_count +
        grade_distributions.f_count) AS total_alunos
FROM courses
JOIN course_offerings ON course_offerings.course_uuid = courses.uuid
JOIN subject_memberships ON subject_memberships.course_offering_uuid = course_offerings.uuid
JOIN subjects ON subjects.code = subject_memberships.subject_code
JOIN sections ON sections.course_offering_uuid = course_offerings.uuid
JOIN grade_distributions ON grade_distributions.course_offering_uuid = course_offerings.uuid
                        AND grade_distributions.section_number = sections.number
WHERE subjects.abbreviation = 'STAT'
GROUP BY courses.uuid, courses.name
HAVING total_alunos > 0
ORDER BY gpa_medio ASC
LIMIT 1

```

Table 4: 1 records

name	gpa_medio	total_alunos
Introduction to the Theory of Probability	2.886374	2231

D) Qual é a disciplina mais fácil?

```

SELECT courses.name,
       SUM(grade_distributions.a_count*4 + grade_distributions.ab_count*3.5 +
           grade_distributions.b_count*3 + grade_distributions.bc_count*2.5 +
           grade_distributions.c_count*2 + grade_distributions.d_count*1) * 1.0 /
       SUM(grade_distributions.a_count + grade_distributions.ab_count +
           grade_distributions.b_count + grade_distributions.bc_count +
           grade_distributions.c_count + grade_distributions.d_count +
           grade_distributions.f_count) AS gpa_medio,
       SUM(grade_distributions.a_count + grade_distributions.ab_count +
           grade_distributions.b_count + grade_distributions.bc_count +
           grade_distributions.c_count + grade_distributions.d_count +
           grade_distributions.f_count) AS total_alunos
FROM courses
JOIN course_offerings ON course_offerings.course_uuid = courses.uuid
JOIN subject_memberships ON subject_memberships.course_offering_uuid = course_offerings.uuid
JOIN subjects ON subjects.code = subject_memberships.subject_code
JOIN sections ON sections.course_offering_uuid = course_offerings.uuid
JOIN grade_distributions ON grade_distributions.course_offering_uuid = course_offerings.uuid
                        AND grade_distributions.section_number = sections.number
WHERE subjects.abbreviation = 'STAT'
GROUP BY courses.uuid, courses.name
HAVING total_alunos > 0
ORDER BY gpa_medio DESC
LIMIT 1

```

Table 5: 1 records

name	gpa_medio	total_alunos
Nonparametric Statistics and Machine Learning Methods	4	24

4. Desconecte do banco de dados.

```
dbDisconnect(conn)
```

```

Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()

```

```
[1] "2026-09-23 19:14:29 -03"
```